Yiyang Huang

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Education

Northeastern University, Boston, MA Sep 2024 – Aug 2029 (Expected)
Ph.D. in Computer Engineering, advised by Prof. Yun Raymond Fu

Xidian University, Xi'an, China Sep 2021 – Jun 2024
M.S. in Computer Science, advised by Prof. Xuefeng Liang

Xidian University, Xi'an, China Sep 2017 – Jun 2021
B.Eng. in Intelligence Science and Technology

Research Interests

Multimodal LLMs and Agents; Agentic Planning and Reflection; Agentic Reinforcement Learning; Tool-Augmented Reasoning; Hallucination Detection and Mitigation; Long-Form Video Understanding; Layout and Document Understanding

Selected Experience

Adobe Research | Vision-Language Lab | Returning Research Intern | San Jose, CA May 2026 – present
Mentors: Dr. Zhaowen Wang, Dr. Shraman Pramanick

Agentic Design Layout Evaluation and Refinement

- Identified limited holistic layout understanding and document reading-order bias as two key challenges in design layout evaluation.
- Developed a **Qwen3-VL-8B-based layout reflection agent** through **GRPO-based reinforcement post-training**, using hierarchical layout understanding and debiased visual reasoning to provide actionable feedback and refinement suggestions, improving the aesthetic quality of designs generated by Codex and Claude Code.

Adobe Research | Vision-Language Lab | Research Intern | San Jose, CA May 2025 – Nov 2025
Mentors: Dr. Zhaowen Wang, Dr. Simon Jenni, Dr. Jing Shi

[ECCV'26; Patent Filed] Compositional Layout Understanding in MLLMs

- Identified semantic drift and structural ambiguity as key challenges in compositional layout reasoning, and developed MASON with metadata-visual grounding and 3D structural perception, achieving 26% and 15% relative improvements in compositional matching over Gemini-2.5-Pro and GPT-5/GPT-o3, respectively, using Qwen2.5-VL-7B as the backbone.
- Constructed CoDeLayout, a ~20K-sample benchmark, by developing an automated pipeline for compositional layout mining and QA generation.

Northeastern University | SMILE Lab | Research Assistant | Boston, MA Sep 2024 – present
Supervisor: Prof. Yun Raymond Fu

[Under Review] Tool-Augmented Agentic Keyframe Selection for Long-Form Video Understanding

- Developed SCOUT, an agentic keyframe selection framework that leverages a lightweight VLM (Qwen2.5-VL-3B) as the planning-and-reflection agent and CLIP as the retrieval tool, boosting the long-form VideoQA accuracy of off-the-shelf VideoLLMs (LLaVA-OneVision, Qwen3-VL, InternVL3, and LLaVA-Video) by 5–10% on MLVU, LVB, and VideoMME.

[ICLR'26] Mitigating Hallucinations in Multimodal LLMs

- Identified encoder-side causes of hallucinations and proposed SHIELD, an inference-time intervention combining token reweighting/subtraction with adversarial contrastive decoding, achieving a 27% relative reduction in captioning hallucinations across multiple MLLM families.

[EMNLP'25] Training-Free Adaptation of Image-Pretrained MLLMs for Video Understanding

- Developed D-CoDe, a plug-and-play framework that adapts image-pretrained MLLMs for VideoQA through question decomposition and dynamic token/frame compression, achieving a 29% relative improvement in egocentric VideoQA accuracy with LLaVA-NeXT-7B.

Publications

First-Author Papers

- **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Jianglin Lu, Qihua Dong, Hailing Wang, Huimin Zeng, Yun Fu. "SCOUT: Selecting Critical Observations via Tiny VLM Planning and Tool-Augmented Retrieval." ***Under review***.
- **Yiyang Huang**, Zhaowen Wang, Simon Jenni, Jing Shi, Yitian Zhang, Yizhou Wang, Yun Fu. "Beyond Atomic Layouts: Compositional Design Understanding with Vision-Language Models." **ECCV 2026**. [Paper] [Code] [Project Page]
- **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Mingyuan Zhang, Liang Shi, Huimin Zeng, Yun Fu. "Distorted or Fabricated? A Survey on Hallucination in Video LLMs." **ACL 2026** (Findings). [Paper] [Repo]
- **Yiyang Huang**, Liang Shi, Yitian Zhang, Yi Xu, Yun Fu. "SHIELD: Suppressing Hallucinations in LVLM Encoders via Bias and Vulnerability Defense." **ICLR 2026**. [Paper] [Code]

- **Yiyang Huang**, Yizhou Wang, Yun Fu. "D-CoDe: Scaling Image-Pretrained VLMs to Video via Dynamic Compression and Question Decomposition." **EMNLP 2025** (Main). [Paper] [Code]
- **Yiyang Huang**, Xuefeng Liang, Chaowei Fang. "CALLip: Lipreading using Contrastive and Attribute Learning." **ACM MM 2021**. [Paper]

Selected Co-Authored Papers

- Hailing Wang, Jianglin Lu, Xu Ma, **Yiyang Huang**, Yun Fu. "StreamVLM: Adapting Offline Vision-Language Models for Streaming Understanding." ***Under review***.
- Mingyuan Zhang, Yue Bai, Yifan Wang, **Yiyang Huang**, Yun Fu. "Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models." ***preprint***. [Paper]
- Shuai Zou, Xuefeng Liang, **Yiyang Huang**. "LipReading for Low-resource Languages by Language Dynamic LoRA." **ICASSP 2025**. [Paper]

Academic Service

Conference Reviewing: NeurIPS; ACL; EMNLP; IEEE Face & Gesture (FG)
Journal Reviewing: ACM TKDD

Teaching

Teaching Assistant

DS 5110: Essentials of Data Science	Fall 2025
DS 5020: Fundamentals of Linear Algebra and Probability	Spring 2026

Skills

Languages: Python, LaTeX
ML Frameworks: PyTorch, Hugging Face Transformers, Hugging Face Datasets
Training/Fine-Tuning Libraries: Accelerate, PEFT, bitsandbytes
RL/Post-Training Frameworks: TRL, verl, OpenRLHF
Inference & Evaluation: vLLM, lmms-eval
Video Processing: ffmpeg, PyAV, decord
Systems/Infrastructure: Linux, Git, Slurm

Awards

National Scholarship, China	2021
Outstanding Student, Xidian University	2022
1st Prize, Undergraduate Computer Design Competition (National Level), China	2021
2nd Prize, RoboMaster National Robotics Competition, China	2019
3rd Prize, ICRA AI Challenge	2019